

Data Structures: Stacks & Queues

Comprehensive Lesson Guide, Detailed Worksheet & Student Practice Set

Subject: Computer Science / Data Structures

Target Audience: Middle / High School Students

Core Concepts: Stacks (LIFO), Queues (FIFO), Operations

Duration: 60 – 90 Minutes

1. Introduction to Data Structures

In daily life, we constantly organize objects to make them easy to find, access, and manage. Think about your school bag, books arranged on a shelf, or lunch tiffin dabbas stacked together. Without organization, finding your homework or lunch becomes stressful and time-consuming.

In computer science, a **Data Structure** is a specialized format for organizing, processing, retrieving, and storing data efficiently. Just like we use different real-world organization methods, computers rely on structured logical patterns to handle information effectively.

2. Deep Dive: Stacks (LIFO Principle)

Key Concept: Last In, First Out (LIFO)

A **Stack** is a linear data structure where elements are added and removed from only one end, known as the **TOP**. The last item placed into the stack is always the very first item to be removed.

Primary Stack Operations

- **Push:** Adds a new element onto the TOP of the stack. (e.g., placing a new plate on top of a stack of plates).
- **Pop:** Removes the top element from the TOP of the stack. You cannot access items lower down without first popping the items above them.
- **Peek / Top (Optional):** Inspects the top item of the stack without removing it.

Real-World Examples of Stacks

- **Wedding Buffet Plates:** Clean plates are stacked vertically; guests take the plate placed last on top.
- **Bangles on a Wrist:** Bangles are put on one by one; the last bangle put on must be pulled off first.
- **Idli Steamer Plates & Tiffin Dabbas:** Stacked vertically; the top lid or container must be lifted first.
- **Browser Back Button:** Web pages you visit are pushed onto a stack; clicking "Back" pops the current page off.

3. Deep Dive: Queues (FIFO Principle)

Key Concept: First In, First Out (FIFO)

A **Queue** is a linear data structure where elements are added at one end called the **REAR** (or Back) and removed from the other end called the **FRONT**. The item that enters first gets served and removed first.

Primary Queue Operations

- **Enqueue:** Adds a new element or person to the REAR (back) of the queue.
- **Dequeue:** Removes an element or person from the FRONT of the queue.
- **Front / Peek:** Reads the item currently at the front of the line without removing it.

Real-World Examples of Queues

- **Railway Ticket Counter Line:** The person who arrives first is served first at the booking window.
- **Auto-Rickshaw Stand Line:** Commuters form a line; the auto at the front takes the passenger who waited longest.
- **Pani Puri Stall Line:** Customers are served pani puris in the exact sequence they arrive.
- **Printer Queue:** Documents sent to a shared printer are printed in the order received.

4. Summary Comparison: Stack vs. Queue

Feature	Stack (LIFO)	Queue (FIFO)
Full Meaning	Last In, First Out	First In, First Out
Insertion Point	TOP (Push)	REAR / BACK (Enqueue)
Deletion Point	TOP (Pop)	FRONT (Dequeue)
Access Structure	Single-ended access point	Double-ended (Front and Rear)
Primary Analogy	Vertical stack of plates / books / bangles	Horizontal waiting line (ticket counter / auto stand)
Fairness Mechanism	Most recent priority (LIFO)	Strict arrival-order fairness (First Come, First Served)

Expanded Interactive Student Worksheet

Guided Exercises, Scenario Classification & Tracing Tasks

Part A: Real-World Scenario Classification & Logic Analysis

Identify whether each scenario represents a Stack (LIFO) or a Queue (FIFO). Explain why by identifying the insertion/deletion points.

1. Metro Train Boarding & Alighting Line

Passengers entering through automatic turnstile gates one by one.

Type & Justification: _____

2. Stack of Hot Chapatis placed on a Dining Plate

Freshly cooked chapatis kept warm under a lid.

Type & Justification: _____

3. Undo/Redo Feature in a Text Editor (e.g., MS Word)

Reverting recent typing errors sequentially.

Type & Justification: _____

Part B: Operation Tracing & Visual State Exercise

Follow the sequence of operations step-by-step and write down the contents of the structure from bottom to top (or front to rear) after each operation.

Exercise B1 (Stack Tracing)

Start with an empty Stack S.

1. Push("Book A") | 2. Push("Book B") | 3. Push("Book C") | 4. Pop() | 5. Push("Book D")

Question: Which book is currently at the TOP of the stack? What are all remaining books from bottom to top?

Answer: _____

Exercise B2 (Queue Tracing)

Start with an empty Queue Q.

1. Enqueue("Rohan") | 2. Enqueue("Priya") | 3. Dequeue() | 4. Enqueue("Aarav") | 5. Enqueue("Sneha")

Question: Who is currently at the FRONT of the queue to be served next? Who is at the REAR?

Answer: _____

Student Practice & Assessment Set

Test Your Knowledge: Stacks vs. Queues (Questions Only)

Instructions: Answer all questions clearly in the space provided. Show your reasoning for conceptual and tracing questions.

Q1. Conceptual Definition

State the full forms of LIFO and FIFO. Explain how the point of entry and point of exit differ between a Stack and a Queue.

Q2. Real-World Scenario Identification

Classify each of the following four scenarios as either **STACK** or **QUEUE**:

- (a) People waiting in line at a Pani Puri stall: _____
- (b) A stack of glass bangles stored inside a cylindrical box: _____
- (c) Songs queued up in a music player playlist: _____
- (d) Plates stacked up in an Idli Steamer container: _____

Q3. Stack Algorithm Execution

An empty Stack S undergoes the following series of operations:

Push(10) → Push(20) → Push(30) → Pop() → Push(40) → Pop() → Push(50)

Tasks:

- 1. What item is popped first? _____
- 2. What item is popped second? _____
- 3. List all elements currently remaining in the stack, starting from Bottom to Top: _____

Q4. Queue Algorithm Execution

An empty Queue Q undergoes the following series of operations:

Enqueue("A") → Enqueue("B") → Dequeue() → Enqueue("C") → Enqueue("D") → Dequeue()

Tasks:

- 1. Which element was removed during the first Dequeue operation? _____
- 2. Which element was removed during the second Dequeue operation? _____
- 3. Which element is currently at the FRONT of the queue? _____
- 4. Which element is currently at the REAR of the queue? _____

Q5. Critical Thinking & Application

Imagine you are designing an online ticket booking platform for a concert. If thousands of users try to book seats simultaneously, would you use a Stack or a Queue to handle ticket requests fairly? Justify your choice based on LIFO vs FIFO principles.
